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(54) **SYSTEM, METHOD, AND COMPUTER PROGRAM FOR PREDICTIVE AUTOSCALING FOR FASTER SEARCHES OF EVENT LOGS IN A CYBERSECURITY SYSTEM**

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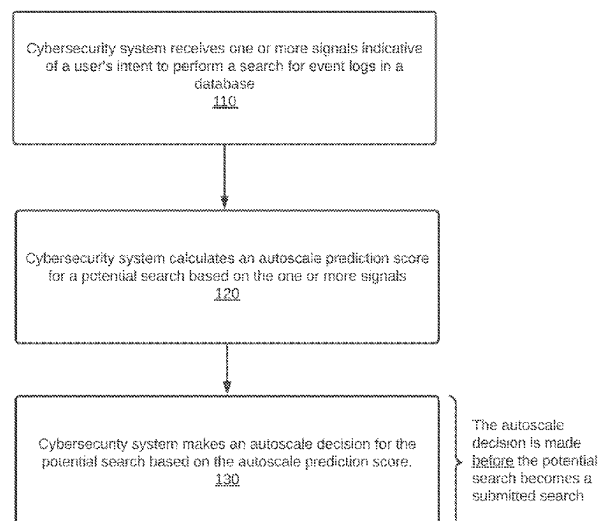
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(57) **ABSTRACT**

The present disclosure describes a system, method, and computer program for predictive autoscaling for faster searches of event logs in a cybersecurity system. In one embodiment, the system receives search-related signals from a plurality of signal sources. The signals are indicative of: (1) a user's intent to perform a search for event logs in a cybersecurity database, (2) how computationally intensive the potential search is likely to be, and (3) the currently available computational resources. The signals are evaluated, and an autoscale prediction score is calculated. The autoscale prediction score reflects the likelihood of a user to submit search, the computational resources required for the potential search, and the currently available computational resources. The system scales computational resources in accordance with the autoscale prediction score. These steps are performed before any search is submitted by the user in a search user interface.

21 Claims, 3 Drawing Sheets

Predictive Autoscaling for Event Log Searches in a Cybersecurity System



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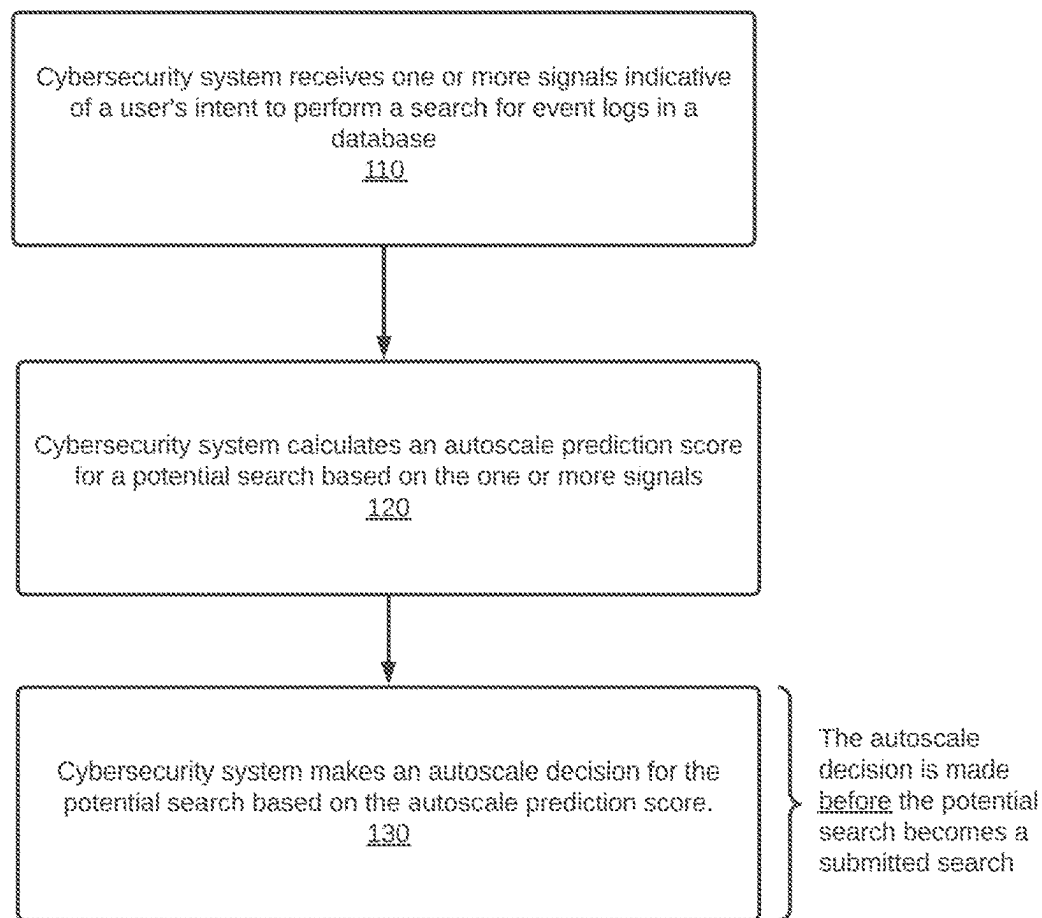
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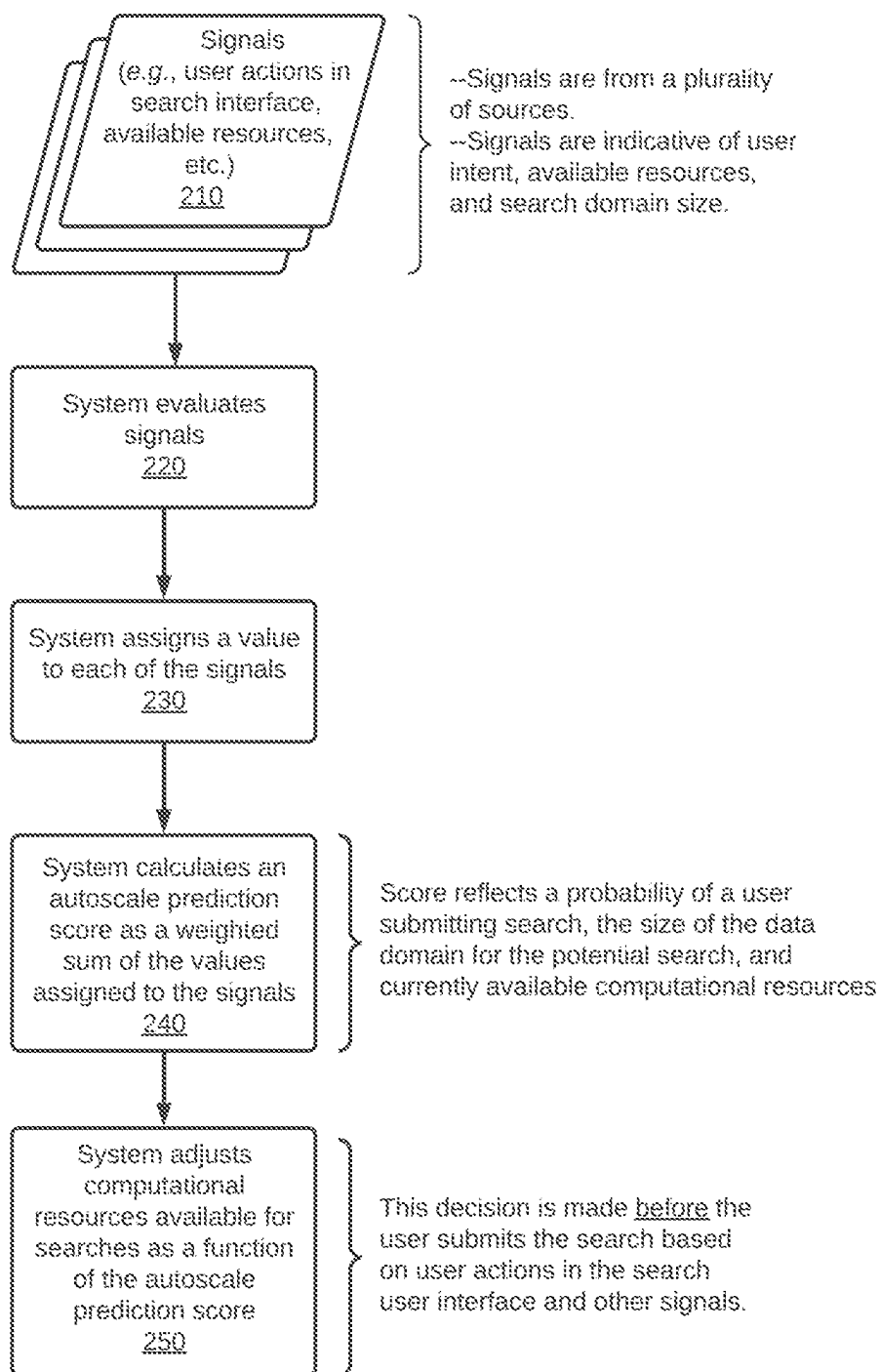
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Predictive Autoscaling for Event Log Searches in a Cybersecurity System

**FIG. 1**

**FIG. 2**

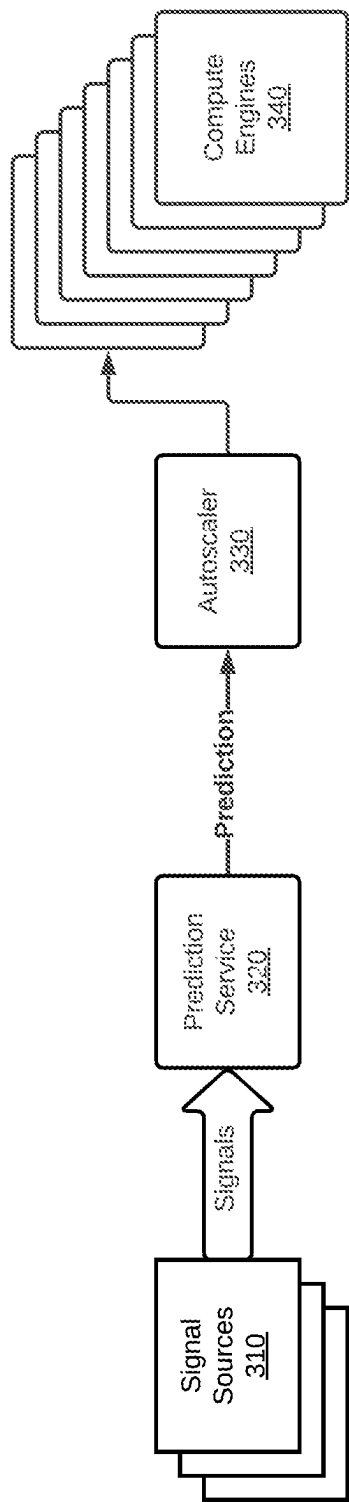


FIG. 3

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SYSTEM, METHOD, AND COMPUTER PROGRAM FOR PREDICTIVE AUTOSCALING FOR FASTER SEARCHES OF EVENT LOGS IN A CYBERSECURITY SYSTEM

BACKGROUND OF THE INVENTION

1. Field of the Invention

This invention relates generally to cybersecurity systems, and, more specifically, to predictive autoscaling for query searches on a database of event logs in a cybersecurity system.

2. Description of the Background Art

Cybersecurity systems monitor entity behavior in a network in order to detect cybersecurity threats. As entities interact with the network, various systems generate raw logs related to the entity behavior. For example, a cybersecurity system may obtain raw data logs related to a user's interactions with the IT infrastructure, such as user logon events, server access events, application access events, and data access events. Cybersecurity system will typically take these raw data logs and generate event logs from the raw data logs.

Cybersecurity analysts within an enterprise perform searches on the event logs to provide a better understanding of cybersecurity risks. Event log databases can be large, and search queries on the event log databases are often compute intensive processes. A query life cycle typically has the following stages:

- Query Authoring
- Query Parsing
- Autoscaling of Compute Resources (horizontal scaling)
- Query Execution
- Query Results
- Downscaling of Compute Resources

Many cybersecurity systems store event logs in cloud-based databases, and scale compute resources for searches on demand. The scaling is performed after the query is submitted and parsed. Dynamically scaling compute resources on demand is more cost efficient than maintaining a high amount of compute resources at all times. Unfortunately, the time required to autoscale compute resources creates a delay in query execution, which results in slow queries. Therefore, there is demand for a solution that can autoscale without slowing down searches.

SUMMARY OF THE DISCLOSURE

The present disclosure describes a system, method, and computer program for predictive autoscaling for faster searches of event logs in a cybersecurity system. The system receives one or more signals indicative of a user's intent to perform a search for event logs in a cybersecurity database. Examples of the signals include a user clicking on a query builder window within a search interface for the event logs database or selecting a previous search. The system calculates an autoscale prediction score for a potential search based on the one or more signals. The system then makes an autoscale decision based on the autoscale prediction score. These steps are all performed prior to a user submitting a search query. In other words, the system makes an autoscale decision based on the likelihood that a user will submit a search. This way, if and when the user does submit a search,

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the computational resources for the search are available and there is no delay associated with provisioning additional compute resources.

In a preferred embodiment, the system receives signals from a plurality of different sources that collectively indicate not only the user's intent to perform a search, but also indicate how computationally intensive the search is likely to be and the currently available computational resources. For example, a signal indicating that a user has selected a query time domain within the search user interface provides an indication of the size of the search domain and hence the amount of computational resources required.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a flowchart that illustrate method for predictive autoscaling for event log searches in accordance with one embodiment.

FIG. 2 is a flowchart that illustrates an example implementation of the method of FIG. 1.

FIG. 3 is a block diagram that illustrates an example architecture for a system for predictive autoscaling.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present disclosure describes a system, method, and computer program for predictive autoscaling for faster searches of event logs in a cybersecurity system. The methods disclosed herein are performed by a computer system ("the system"), such as cybersecurity system that detects cybersecurity threats in a network.

FIG. 1 illustrates a method for predictive autoscaling for event log searches in accordance with one embodiment. The system receives one or more signals indicative of a user's intent to perform a search for event logs in a cybersecurity database (step 110). Examples of the signals include a user clicking on a query builder window within a search interface for the event logs database or selecting a previous search. The system calculates an autoscale prediction score for a potential search based on the one or more signals (step 120). The system then makes an autoscale decision based on the autoscale prediction score (step 130). These steps are all performed prior to a user submitting a search query. In other words, the system makes an autoscale decision based on the likelihood that a user will submit an event log search. This way, if and when the user does submit a search, the computational resources for the search are available, and there is no delay associated with provisioning additional compute resources.

FIG. 2 illustrates an example implementation of the method of FIG. 1. The system receives signals from a plurality of sources that are collectively indicative of: (1) user intent to perform a search, (2) likely search domain size, and (3) currently available computational resources (step 210). The signals related to the user intent and to the search domain size include signals about a user's actions on a search user interface before the user submits a search. Examples of the signals are set forth below in Table 1.

The system evaluates the signals and assigns a value to each signal (steps 220-230). The table below lists examples of the signals received and the values which may be assigned to each signal:

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TABLE 1

Signal	What the signal is indicative of	Example Range of Values
1 User is using a query builder on a search user interface (e.g., user clicks on a query bar).	User intent to search	0 (no usage), 1 (usage)
2 User selects a query time window (e.g., time range selection)	User intent to search; Size of search domain (i.e., computational resources required for search)	0 (no selection of a window), 0.1 (selection of a "small" window), 0.2 (selection of a "medium" window), and 0.3 (selection of a "large" window). Small, medium, and large are defined by time window ranges.
3 User selection of a recent query	User intent	0 (no selection), 1 (selection)
4 Pattern of usage hours for the given user	User intent	0 (current time is outside the user's peak search timeframe), 1 (current time is within the user's peak search timeframe)
5 Pattern of usage hours for the given customer entity	User intent	0 (current time is outside the entity's peak usage window for searches), 1 (current time is within the entity's peak usage window for searches)
6 Adding indexed fields to the query builder	User intent to search and computational resources required	0 (an index field is not yet included within the search criteria), 1 (an indexed field has been added to the search criteria)
7 Percentage of compute resources available	Available compute resources	0.3 (no compute engines currently available), 0.2 (small amount of compute engines available), 0.1 (medium amount of compute engines available), 0 (large amount of compute engines available). Small, medium, and large values may be predefined in configuring the system.
8 Size of customer event log database	Size of search domain	0.1 (small), 0.2 (medium), 0.3 (large). Small, medium, and large values are defined by data size ranges.

The system calculates an autoscale prediction score as a weighted sum of the values assigned to the signals (step 240). The score reflects the probability of a user submitting a search, the size of the search domain, and currently available computational resources. The autoscale prediction score may be expressed mathematically as follows:

$$\text{Autoscale prediction score}(APS) = s1*w1 + s2*w2 + s3*w3 + s4*w4 + \dots sN*wN$$

Where N is the number of signals.

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In one embodiment, the signal values and the weights are assigned such that the autoscale prediction score is in the range between 0 and 1. The example signal values in Table 1 are designed for a prediction score between 0 and 1.

The system adjusts computational resources available for searches as a function of the autoscale prediction score (step 250). The autoscale prediction score is used to decide whether to autoscale and the capacity desired. This may be expressed mathematically as followed:

10 If $APS > \text{threshold value}$, Autoscaler initiates autoscaling.
Capacity required = $f(APS)$

$f(APS)$ translates the autoscale prediction score to the amount of compute resources to be provisioned by the Autoscaler.

15 Steps 210-250 are performed before a user submits a search.

FIG. 3 illustrates an example architecture of the system. A Prediction Service module 320 receives signals from a plurality of signal sources 310. The Prediction Service Module 320 calculates an autoscale prediction score in accordance with the methods described herein. The Autoscaler 330 provisions compute engines 340 for searching as a function of the autoscale prediction score.

The methods described with respect to FIGS. 1-3 are embodied in software and performed by a computer system (comprising one or more computing devices) executing the software. A person skilled in the art would understand that a computer system has one or more memory units, disks, or other physical, computer-readable storage media for storing software instructions, as well as one or more processors for executing the software instructions.

As will be understood by those familiar with the art, the invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. Accordingly, the above disclosure is intended to be illustrative, but not limiting, of the scope of the invention.

The invention claimed is:

1. A non-transitory computer-readable medium comprising a computer program, that, when executed by a computer system, enables the computer system to perform the following method for predicting autoscaling requirements for query searches on a database of event logs in a cybersecurity system, the method comprising:

45 receiving a signal indicative of a user's intent to perform a search for event logs in the cybersecurity system;
calculating an autoscale prediction score for a potential search based on the signal; and
making an autoscale decision for the potential search based on the autoscale prediction score, wherein the autoscale decision for the potential search is made before the potential search becomes a submitted search for event logs in the cybersecurity system, and wherein the autoscale decision is to make computational resources available for the potential search, if and when the potential search becomes a submitted search.

2. The non-transitory computer-readable medium of claim 1, wherein:

60 a plurality of signals is received from a plurality of different signal sources for use in calculating the autoscale prediction score;

the plurality of signals includes the signal indicative of the user's intent to submit a query for event logs and a signal indicative of currently available computational resources for performing searches; and

the autoscale prediction score is calculated based on the plurality of signals.

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3. The non-transitory computer-readable medium of claim 2, wherein the plurality of signals also includes a signal indicative of computational resources required for the potential search.

4. The non-transitory computer-readable medium of claim 3, wherein the signal indicative of the computational resources required for the potential search is one of the following: selection of a query time domain, size of event log database upon which the potential search would be performed, and addition of an indexed field to a query builder.

5. The non-transitory computer-readable medium of claim 2, wherein the signal indicative of the user's intent to perform the search is one of the following: usage of a query builder for the event log search, selection of a recent query, the user's past patterns of searches, selection of a query time domain, and an addition of an indexed field to a query builder.

6. The non-transitory computer-readable medium of claim 2, wherein calculating the autoscale prediction score comprises:

assigning a value to each of the plurality of signals; and calculating a weighted sum of the signal values.

7. The non-transitory computer-readable medium of claim 1, further comprising:

receiving a new signal related to the potential search after the autoscale prediction score has been calculated for the potential search;

recalculating the autoscale prediction score to factor in the new signal; and

making a new autoscale decision based on the recalculated autoscale prediction score.

8. The non-transitory computer-readable medium of claim 1, wherein making an autoscale decision comprises adjusting computational resources available for the potential search.

9. The non-transitory computer-readable medium of claim 8, wherein computational resources are increased as a function of the autoscale prediction score.

10. A computer system for predicting autoscaling requirements for query searches on a database of event logs in a cybersecurity system, the system comprising:

one or more processors;

one or more memory units coupled to the one or more processors, wherein the one or more memory units store instructions that, when executed by the one or more processors, cause the system to perform the operations of:

receiving a signal indicative of a user's intent to perform a search for event logs in the cybersecurity system;

calculating an autoscale prediction score for a potential search based on the signal; and

making an autoscale decision for the potential search based on the autoscale prediction score, wherein the autoscale decision for the potential search is made before the potential search becomes a submitted search for event logs in the cybersecurity system, and wherein the autoscale decision is to make computational resources available for the potential search, if and when the potential search becomes a submitted search.

11. The system of claim 10, wherein:

a plurality of signals is received from a plurality of different signal sources for use in calculating the autoscale prediction score;

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the plurality of signals includes the signal indicative of the user's intent to submit a query for event logs and a signal indicative of currently available computational resources for performing searches; and

the autoscale prediction score is calculated based on the plurality of signals.

12. The system of claim 11, wherein the plurality of signals also includes a signal indicative of computational resources required for the potential search.

13. The system of claim 12, wherein the signal indicative of the computational resources required for the potential search is one of the following: selection of a query time domain, size of event log database upon which the potential search would be performed, and addition of an indexed field to a query builder.

14. The system of claim 11, wherein the signal indicative of the user's intent to perform the search is one of the following: usage of a query builder for the event log search, selection of a recent query, the user's past patterns of searches, selection of a query time domain, and an addition of an indexed field to a query builder.

15. The system of claim 11, wherein calculating the autoscale prediction score comprises:

assigning a value to each of the plurality of signals; and calculating a weighted sum of the signal values.

16. The system of claim 10, further comprising:

receiving a new signal related to the potential search after the autoscale prediction score has been calculated for the potential search;

recalculating the autoscale prediction score to factor in the new signal; and

making a new autoscale decision based on the recalculated autoscale prediction score.

17. The system of claim 10, wherein making an autoscale decision comprises adjusting computational resources available for the potential search.

18. The system of claim 17, wherein computational resources are increased as a function of the autoscale prediction score.

19. A method, performed by a computer system, for predicting autoscaling requirements for query searches on a database of event logs in a cybersecurity system, the method comprising:

receiving a signal indicative of a user's intent to perform a search for event logs in the cybersecurity system;

calculating an autoscale prediction score for a potential search based on the signal; and

making an autoscale decision for the potential search based on the autoscale prediction score, wherein the autoscale decision for the potential search is made before the potential search becomes a submitted search for event logs in the cybersecurity system, and wherein the autoscale decision is to make computational resources available for the potential search, if and when the potential search becomes a submitted search.

20. The method of claim 19, wherein:

a plurality of signals is received from a plurality of different signal sources for use in calculating the autoscale prediction score;

the plurality of signals includes the signal indicative of the user's intent to submit a query for event logs and a signal indicative of currently available computational resources for performing searches; and

the autoscale prediction score is calculated based on the plurality of signals.

21. The method of claim **20**, wherein the plurality of signals also includes a signal indicative of computational resources required for the potential search.

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